

FSOC Tracker: User Manual

AI-based virtual camera tracking system for coarse alignment of mobile Free Space Optical Communication (FSOC) terminals. Smart India Hackathon problem statement SIH26169, Department of Space / ISRO Space Applications Centre.

1. What the application does

The application simulates the coarse alignment stage of an FSOC link entirely in software.

- It draws a scene (the "screen", at least 2000 x 2000 pixels) with a moving optical beacon.
- A virtual pan-tilt camera looks at a 640 x 480 pixel window of that scene (4 x 3 degrees) and can turn at a limited speed, like a real motorised mount.
- Disturbances can be added: fog, haze, rain, low light, atmospheric turbulence, platform motion, camera vibration, and sensor noise.
- The tracker detects the beacon, measures its centre to a fraction of a pixel, predicts where it is going, and steers the camera window to keep it centred.
- Every run writes a per-frame log (`frames.csv`), a summary (`summary.json`) and an automatic performance report (`report.pdf`).
- For Benchmark 2, an `.mp4` video can be opened in place of the simulated scene.

2. Installation

2.1 Standalone executable (no Python needed)

1. Download the archive for your operating system from the release folder (`FSOC-Tracker-windows.zip`, `FSOC-Tracker-linux.tar.gz`, `FSOC-Tracker-macos.zip`).
2. Extract it anywhere.
3. Run `FSOC-Tracker` (Windows: `FSOC-Tracker.exe`). The first start takes a few seconds.

Results are written to a `results` folder next to the executable.

2.2 From source

Requires Python 3.11 or 3.12.

```
git clone <repository>
cd SIH169
python -m venv .venv
.venv/bin/pip install -e ".[dev]"          # Windows: .venv\Scripts\pip install -e ".[dev]"
.venv/bin/fsoc-tracker-gui                # start the desktop application
```

Optional, to retrain the AI detector: `pip install -e ".[train]"` then python `training/train_heatmap.py`.

3. The main window

```
+-----+-----+-----+
--+
| Parameters      | State | Acquisition | Track err | Centroid | Lock | FPS |
Telemetry |
| (scrollable,    +-----+ one value per
| hover for help) | Screen view          | Camera view          | line, every
|                  | whole scene, camera  | 640 x 480 with HUD   | frame
|                  | window, trails, 2 deg | capture ring, error |
|                  | grid, legend         | vector, 1 deg bar    |
|                  +-----+
|                  | Errors      | Gimbal command      |
|                  | Timing      | Tracker state        |
+-----+-----+-----+
--+
| Status: what is running and where results go          [=====]
|
+-----+-----+-----+
+
```

Toolbar, left to right

Control	Action
Scenario	Pick a ready-made test case from <code>configs/scenarios</code> . "Custom" keeps whatever is in the panel.
Start / Pause / Step / Stop	Run control. Space starts or pauses, N steps one frame while paused, Esc stops.
Speed	0.25x to 4x real time, or Max speed (shows the true processing rate).
Open video (Benchmark 2)	Choose an <code>.mp4</code> ; the simulator is bypassed and the video frames become the scene. Ctrl+O.
Use simulator	Return to the simulated scene.
Save scenario	Save the panel as a <code>.yaml</code> ; it appears in the Scenario list. Ctrl+S.
Screenshot	Save a PNG of the window into <code>results/</code> . Ctrl+P.
Results folder	Open <code>results/</code> .

Control	Action
User manual	Open this manual.
Help	Legend and output summary.

Tiles (live specification check)

Six tiles above the pictures update every frame and turn green when the problem statement specification is met, amber when it is not: tracker state, acquisition time (2 s or less), mean tracking error since acquisition (10 px or less), mean centroiding error, lock retention (target loss under 5 percent), and processing FPS (20 or more). In video mode the two error tiles read "n/a" because the video carries no ground truth.

Screen view (left picture)

The whole scene, downscaled, with a 2 degree grid. Cyan rectangle: the camera window and its centre. Orange circle: the true beacon (simulator only). Green cross: the tracker's estimate. Grey circles: other targets. Faint trails: where the beacon and the window have been. The legend is printed along the bottom edge.

Camera view (right picture)

What the camera window sees, contrast-stretched for display. Cyan cross: boresight. Dashed ring: the capture radius, green when locked. Line from the centre: the pointing error vector. Green box: this frame's detection. Orange dot: the prediction, with a fainter ring for the estimator's uncertainty. Green trail: recent estimates. A 1 degree scale bar sits bottom right. The banner shows the state, tracking error in pixels and arcseconds, confidence and which detector produced the measurement; the bottom line shows pan, tilt, the commanded rates, saturation and processing time.

Plots

- Errors: tracking error (window centre to beacon, cyan) and centroiding error (measured centre to true centre, orange). Dashed line: the 10 pixel specification.
- Gimbal command: pan and tilt rate commands with the rate limit dashed.
- Processing time per frame with the 20 FPS budget (50 ms) dashed.
- Tracker state: SEARCH, VERIFY, TRACK, COAST, REACQUIRE.

Telemetry column

Every internal quantity for the current frame: state, detector, candidates, confidence, SNR, pose, commands, detection, estimate, velocity, uncertainty, motion model probabilities, truth, errors and timing. Before a run it shows a short how-to.

4. Parameters

Every row of the problem statement parameter table has a control. Values outside the suggested range are allowed where the table says "user-defined".

Run

- Name, Seed: the seed makes a run exactly repeatable.
- Duration (s): length of a simulator run. Ignored for video input (the whole video runs).
- Extra targets: number of additional beacons (decoys) added with random paths.

Screen (rows 1 to 2)

- Width, Height: scene size in pixels. Default 2000 x 2000.
- Colour camera: renders a three-channel colour frame (PS row 2, optional); the tracker always works on luminance, so results are identical.
- Background: starfield, terrain, gradient or flat. Sky level and star density.

Camera (rows 3 to 6, 13 to 15)

- Width, Height: camera window in pixels. Default 640 x 480.
- FOV width, height: degrees. Default 4 x 3. Degrees per pixel is derived from these.
- Update rate: frames per second. Default 30.
- Max pan, Max tilt: degrees per second. Default 5.
- Max accel, Command latency: gimbal realism.
- Hard mode: the tracker sees only the pixels inside the window and must search.

Designated target (rows 7 to 12)

- Shape: square (default), circle, gaussian. Size in pixels (default 10). Peak intensity.
- Motion: line, circular, figure8, random, spiral, sinusoidal, static.
- Speed, Radius, Period, Heading: path parameters. Start: random or centre.
- Blink: optional intensity modulation in Hz. 0 is steady.

Disturbances (rows 21 to 25)

- Salt and pepper fraction (0.10 = 10 percent), Gaussian sigma (up to 20), Poisson.
- Camera jitter: pixels per frame, up to 20.
- Atmosphere preset: clear, haze, fog, rain, lowlight. Selecting a preset fills contrast, brightness, blur and turbulence; each can then be edited.
- Platform motion: none, linear, circular, random, spiral, figure8; speed in pixels per frame (up to 20) and period.

Tracker

- Detector: hybrid (classical first, AI fills gaps), classical, cnn.
- Use picture-shift estimate: phase correlation as a vibration hint.
- Controller gains, deadband, capture radius, estimator lag, minimum confidence to acquire.
- Hard mode (Camera section) restricts the tracker to the window; SEARCH then flies an expanding square spiral of window-sized cells at the rate limit. A full sweep of a 2000 px screen at 5 deg/s takes about 12 s, so acquisition in hard mode is 3 to 12 s depending on where the beacon is.

5. Running Benchmark 2 (video input)

1. Click "Open video (Benchmark 2)" and choose the `.mp4` file.
2. Set the camera window size and FOV if the graders specify them.
3. Click Start. The video frames are used as the scene; nothing is drawn by the simulator.
4. When the run ends, `frames.csv` contains the measured beacon centre for every frame (`det_x`, `det_y`) and `report.pdf` contains acquisition time, re-acquisition time, lock retention rate and FPS. Tracking and centroiding error against truth are not available because the video carries no ground truth.

From the command line: `fsoc-tracker video path/to/file.mp4`

6. Scenario files and the command line

A scenario is a YAML file with the same fields as the parameter panel. Example:

```
name: fog_circular
seed: 6
duration_s: 30
targets:
  - {shape: square, size_px: 10, motion: circular, radius_px: 400, period_s: 15}
disturbance:
  atmosphere: fog
  gaussian_sigma: 8
```

Commands:

```
fsoc-tracker run --scenario configs/scenarios/fog_circular.yaml [--seed 3] [--
duration 20]
fsoc-tracker video path/to/file.mp4
fsoc-tracker batch --scenario configs/scenarios/*.yaml --seeds 0-49
fsoc-tracker gui
```

`batch` runs every scenario over every seed and writes `envelope.md`, a table of the worst and mean results per scenario.

7. Output files

Each run creates `results/<name>_<timestamp>/`:

File	Contents
<code>frames.csv</code>	One row per frame: time, state, detection, estimate, camera pose, commands, truth, errors, processing time.
<code>summary.json</code>	All metrics with their definitions and pass/fail against the specification.

File	Contents
<code>report.pdf</code>	Three pages: specification check and metric table; time series; paths, histograms and model probabilities.
<code>scenario.yaml</code>	The exact parameters used, so the run can be repeated.

8. Metric definitions

Metric	Definition
Acquisition time	First frame in TRACK with the tracked beacon within the capture radius (30 px) of the window centre. Spec: 2 s or less.
Tracking error	Distance from the true beacon to the window centre, over frames after acquisition. Spec: 10 px or less (mean).
Tracking error, vibration removed	The same with the per-frame camera vibration subtracted from the truth: the part a rate-limited gimbal can physically follow.
Centroiding error	Distance from the measured beacon centre to the true centre.
Lock retention	Percentage of frames after acquisition in TRACK with the beacon inside the capture radius. Target loss is 100 minus this. Spec: loss under 5 percent.
Re-acquisition time	Time from losing lock to regaining it. Spec: 1 s or less.
FPS	1 divided by per-frame processing time, averaged. Spec: 20 or more.

9. Troubleshooting

- The window does not start on Linux: install `libxcb-cursor0` (Qt 6 requirement).
- Low frame rate: reduce the screen size, disable Poisson noise, or uncheck Real-time pacing to see the true processing speed.
- "Cannot open video": the file must be readable by OpenCV (H.264 `.mp4` is safest).
- The AI detector is not used: `models/beacon_heatmap.onnx` is missing; the classical detector runs alone. Retrain with `python training/train_heatmap.py`.